UNIVERSITY OF LIVERPOOL

01

ENGG320 Product Development

Doozi

2022 · Teamwork (2 members) · University of Liverpool

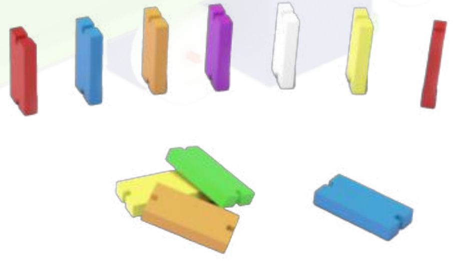
This topic discusses how to help children with disabilities construct a healthy environment for intellectual and psychological growth.

Doozi is a set of wheelchair accessories for children with disabilities which has two important functional components, a domino cart for multi-skill learning and entertainment for children with disabilities, and a musical puzzle carpet for interactive entertainment with friends.

Contribution: Research & Drawing & Modelling

Supervisor: Nickpour, Farnaz

Supervisor's email: Farnaz.Nickpour@liverpool.ac.uk



DESIGN PROCESS

Market Research → Children's Wheelchairs

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User Define → User | Purchaser

↓

User Requirement

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Design Brief → Scenario | PDS

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Concept → Brainstorm | Ideation | Development (Prototype)

↓

Working Principle

↓

Self-Reflection

Market research

The Children's Wheelchairs is designed to give children who are unable to move more experiences of interacting autonomously with their surroundings or people and gaining different stimuli through their own movement. It is not just a substitute for crawling or walking, it is a learning tool that helps children to move and explore and discover. Bugzi and Wizzybug are two of the more common brands of children's wheelchairs on the market.

Wizzybug Age 1-5 Used Indoors & Outdoors

Ergonomic chair

With Wizzybug's specialized seating system, children with higher postural requirements can be provided with a full tray for upper body support as well as different head support options.

Electrical leakage

The product must not be used in areas where water is present, otherwise there is a risk of leakage.

Heavy and inconvenient to transport

The heavy weight of the product and the fact that it cannot be easily dismantled because of the electronic equipment, such as the battery, make it problematic to transport the product.

Bugzi Age 1-5 Used Indoors

Flexzi

The Flexzi allows small, light objects (such as switches and other items) to be easily placed in a position that is accessible to a person.

Switches

A child can operate the Bugzi using a variety of switches. The switches are not required to lie flat on the tray in front of the child to function.

Limited usage scenarios

Can only be used indoors on a flat surface and almost impossible to cross steps.

User Define

Though the comparison we chose to improve Bugzi

Jackson

4 Year-old
An disable child
User

Fustration

He is unable to play enjoyably with his friends while using a wheelchair

The sympathetic glances of his friends often made Jackson feel uncomfortable

Goals

Can play with friends normally

Want to be a wheelchair user who is envied by his friends

Activity Area


Home


Kindergarten


Play House

Jenny

31 Year-old
Have a 3 year-old disable child
Purchaser

Fustration

The amount of time and energy required to care for the children makes her feel very tired

Worried that her child will not be well integrated into the circle of other people her age

Goals

Her child is able to play with peers without the limitations of the Bugzi

Can develop games adapted to children in Bugzi

User Experience


Pedals


Pushbuttons


Building Blocks

" **How might we** design a safe and healthy learning and recreational environment for children with disabilities "

User Requirement

Wishes

44%
Technology
Using external high-tech technology to enhance the enjoyment, usability and convenience of products

52%
Colour
The colours of the product are rich and vibrant.

25%
Sound
Featuring a wealth of sound as a means of entertainment.

40%
Control
There are more personalised controls.

38%
Food & Drink
The wheelchair can be used to store and make food and share it with friends.

35%
Storage
The wheelchair has a storage space for personal belongings.

36%
Relationship
Features that make wheelchairs work for both others and users, and disability commentary around the way users interact with or compare themselves to others.

25%
Disability Remark
People with disabilities will not be discriminated against or pitied for using a wheelchair.

Demands

Price
A reasonable price should be charged for the product.

Safety
The product must not cause harm to the Bugzi user or to the wheelchair itself.

Durability
Wear and fall resistance for this product should be good.

Environmental
It should be suitable for use in its intended indoor environment.
The material of the product should be environmentally friendly.

Compatibility
The product should be in line with the current Bugzi's aesthetic.
The connection method of the product should be robust, easy and intuitive.

Entertaining
A product should be fun for children aged 3-5 and have the potential to foster new ways of playing.

Design Brief

WHO
Bugzi Users
Friends

WHY
Increase Inclusive
Enhancing Fun

WHAT
Providing an outlet for including and integrating disabled children into children's play.

WHERE
Home
Kindergarten & Playhouse

WHEN
Playtime

HOW
Research material must be gathered and an analysis of what wheelchair children really want conducted in order to better understand their developmental needs and psychological state.

User Scenario

A child playing a game with Bugzi

Many children are attracted to join him and play with him

Product Development Specification

Aspect	Objective	Criteria
Performance	Playability	The product should allow children to discover many different ways of playing on their own.
	Attachment	The product as attachment should be easy to mount and remove.
	Cooperation Play	The product should be consider fun by the Bugzi user and their peers.
Cost	Suitable	The production cost should be in the range of 60%-70% of the cost to consumer.
Aesthetics	Attractive styling	The product should be designed in keeping with the preferences of children aged 3-5.
	Match with Bugzi	Product design should match Bugzi's design style.
Safety	High	The product must comply with the European Toy Safety Directive 2009/48/EC
		The product should be designed to avoid misuse and harm.
Product Life Span	Maximise	The product should have a service life of greater than 5 years.
Environment	Indoor	The product should be able to freely move around in an indoor environment (carpet, wooden floor, etc.).
	Room Temperature	The product should be able to perform properly from -10 to 35 degrees Celsius.
	Wear	The product should be hard-wearing.
	Collision	The product should be able to withstand the impact that Bugzi may generate during operation.
Size	Fit with Bugzi	The dimensions of the product should be compatible with Bugzi for easy assembly. l*w*h <= 80*40*60
Weight	Low	The weight of the product should be less than 5kg.
Mantainance	Easy	The product should be easily repairable.
Material	Appropriate to design	The material of the product should meet the specific needs of the design (hard-wearing, impact resistant, etc.).
Ergonomics	Match arm	The product should be sized to fit the child's arm length so that the child can handle it independently.
	Comfortable to use	The product should be ergonomically correct for children ages 3-5, so that they can use it comfortably.
Customer	Emotional connection	The product should be interactive with the user so that they can create emotional connections with the product.
	Control	The user should have full control of the product so that they feel safe when using it.

Brainstorm

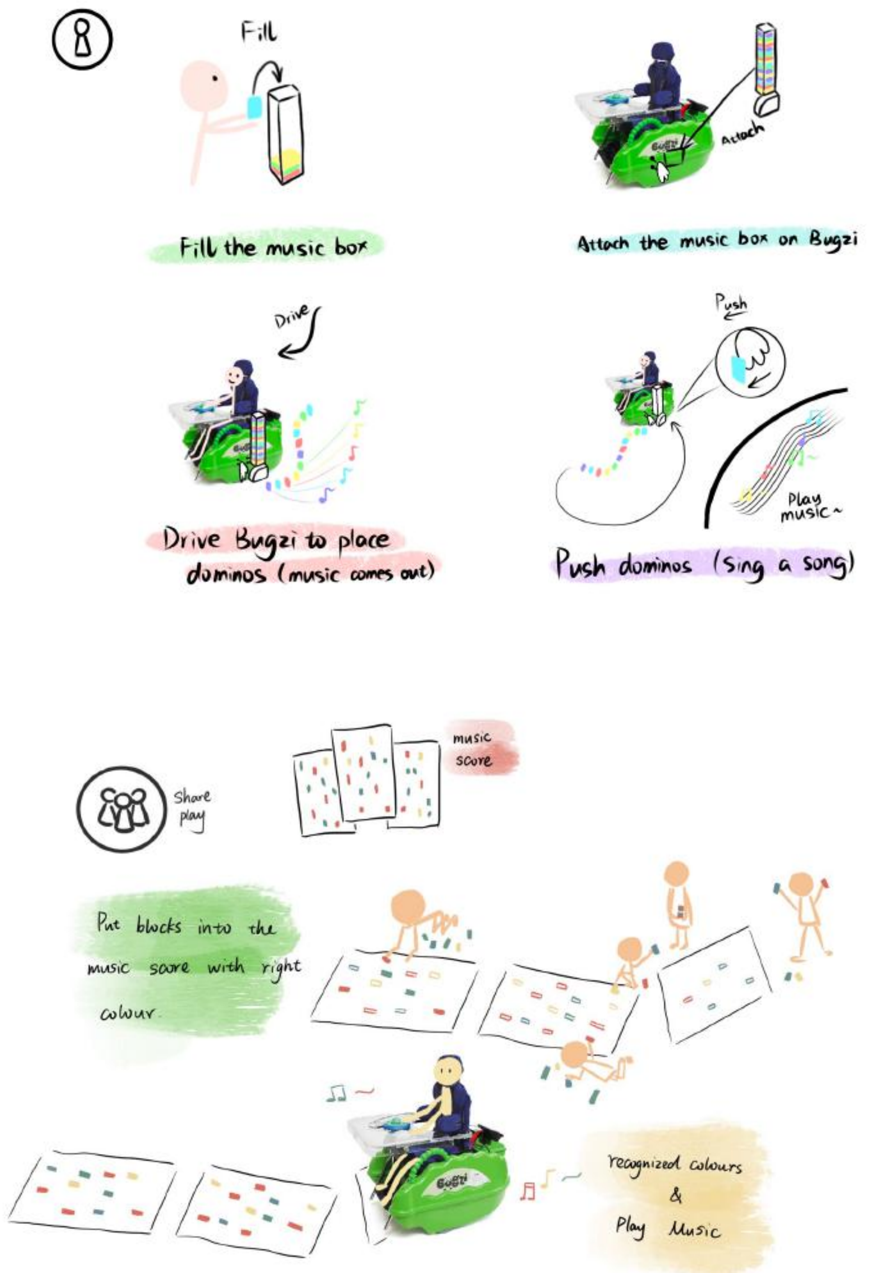
Have Fun & Learn

Share Play

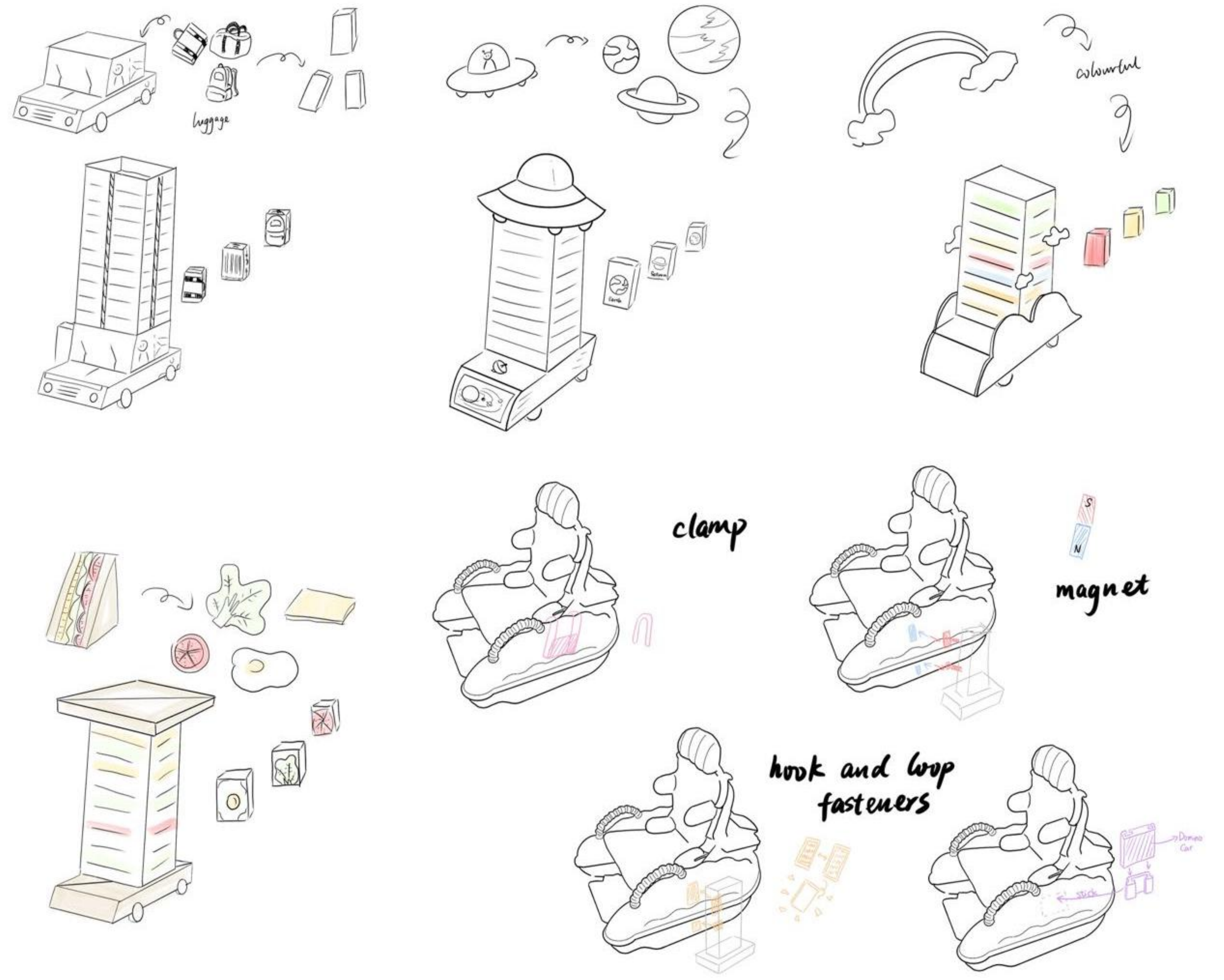
Increase self-esteem

Feel equal in play

Ideation

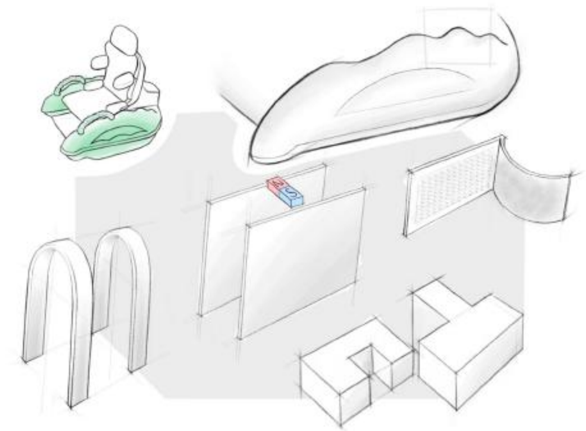


Generation

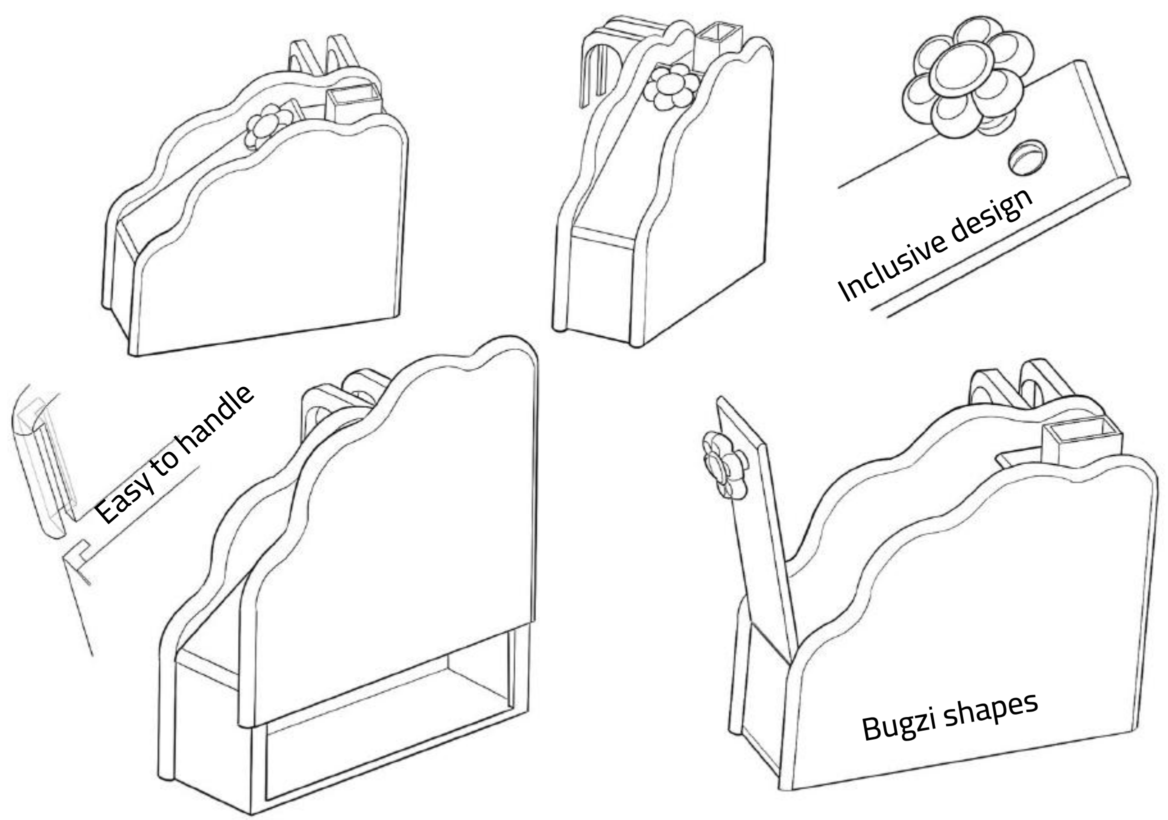


Development

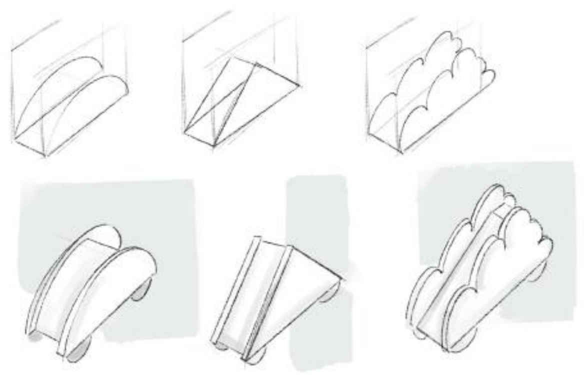
Attachment method



Final sketch



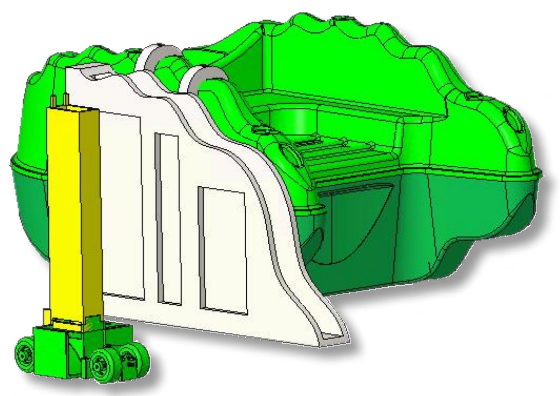
Appearance



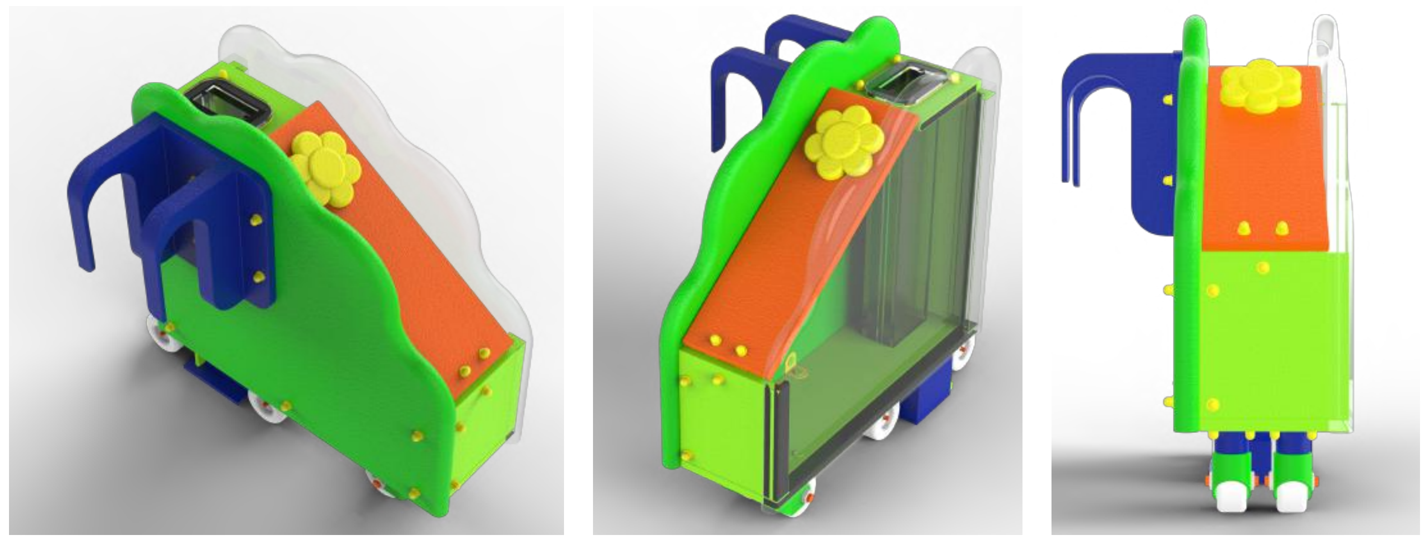
Using paper and foam to make a model to determind the dimension of the box and the car.

Prototype

CAD rough model to test the outfit



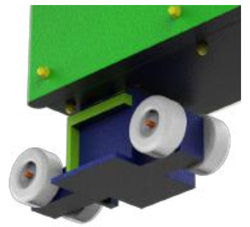
Final CAD model



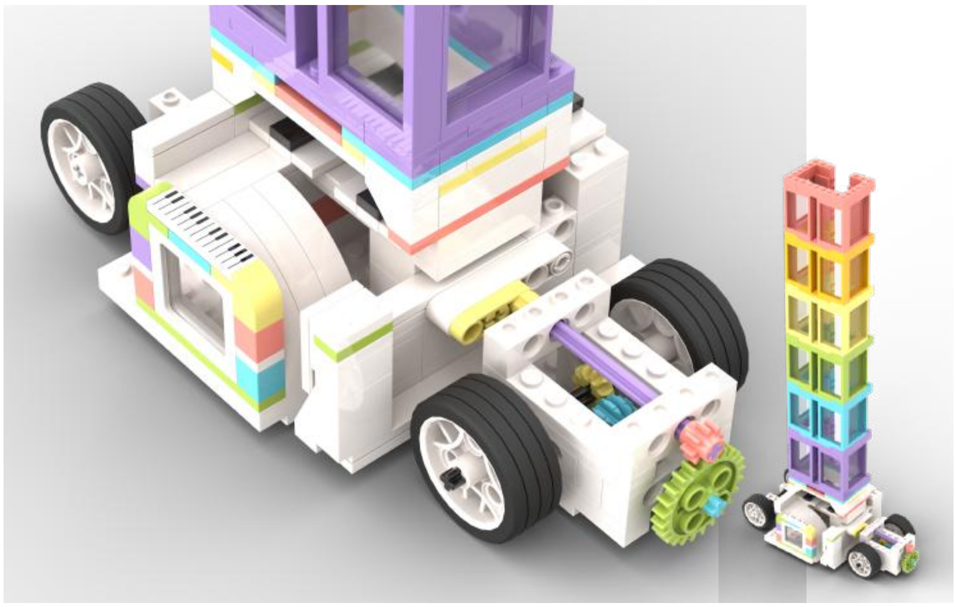
Detail

Plastic nuts

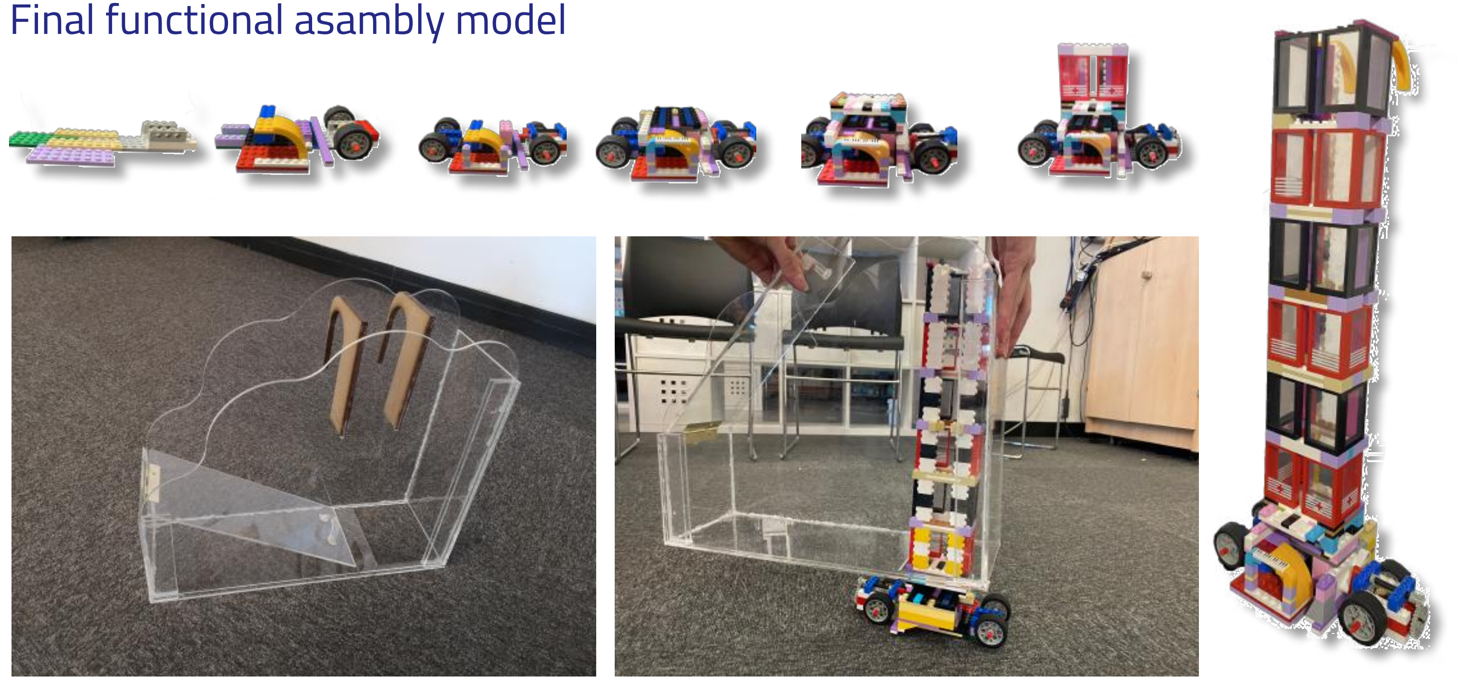
Bottom



Rendered functional model



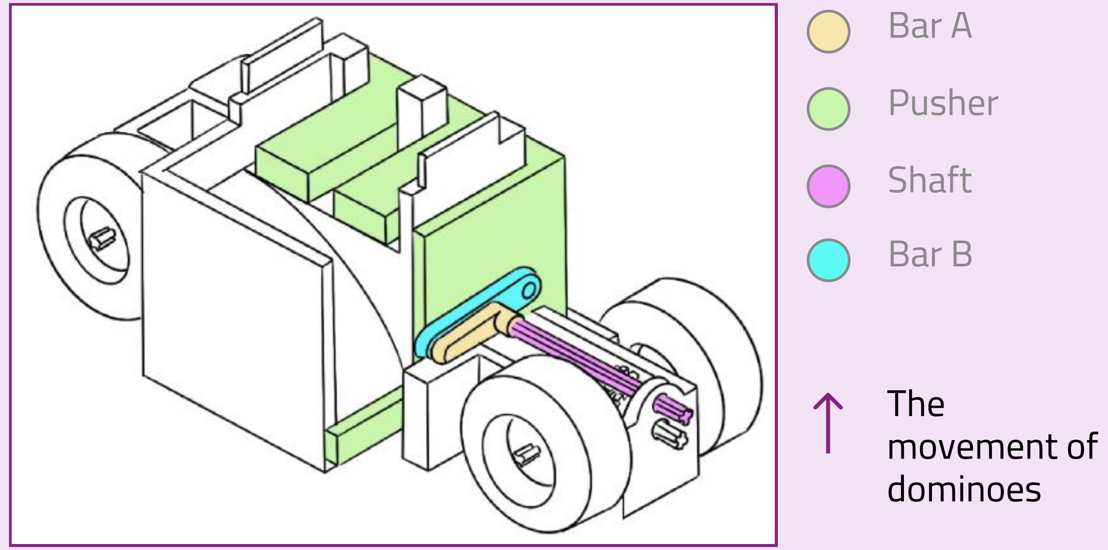
Final functional asambly model



Working Principle

Principle of mechanics

- The Doozi car uses the Bugzi's wheels to rotate the shafts to arrange the dominoes.



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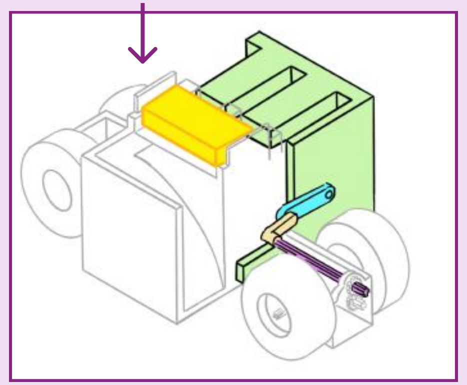
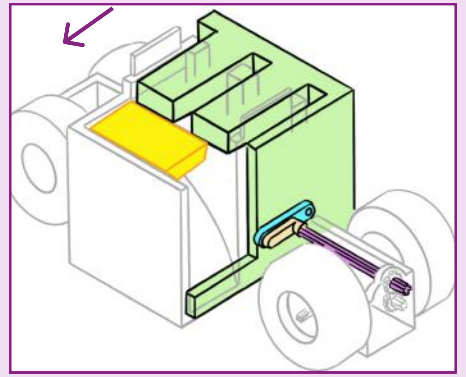


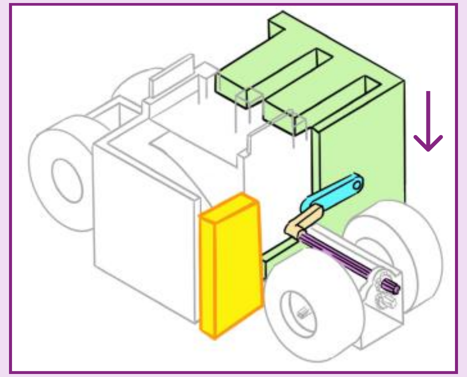
Domino **drops** onto the base.

02



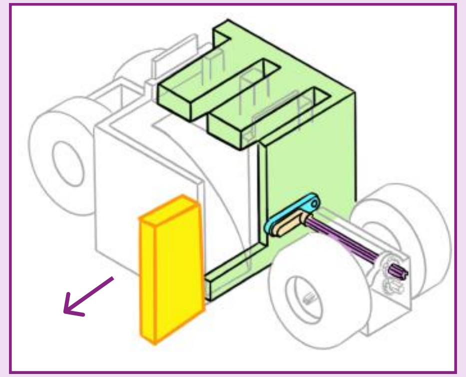
When the product moves forward, the BarA rotates and drives ShaftA and ShaftB which pushes the Pusher **forward**.

03



The Pusher push the domino. The domino **slides down**.

04



The Pusher push the domino again. This domino is **pushed out** and joins the queue.

Principle of sound generation

Converting color codes to tone codes using the C programming language

	Wave length		Scale		Frequency
Red	625-740nm	~	1	~	264Hz
Orange	590-625nm	~	2	~	297Hz
Yellow	565-590nm	~	3	~	330Hz
Green	500-565nm	~	4	~	352Hz
Blue	440-500nm	~	5	~	396Hz
Purple	380-440nm	~	6	~	440Hz

- The users can stack the dominoes in the Doozi car as they desire.
- Participants place dominoes on a carpet puzzle in the order shown on the music sheet.